

PROJECT IMPROVEMENT REPORT - Test 2

Book Management System - Laravel Application

Wambede Ibrahim - MAY23/BSE/3167U

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1. PROJECT OVERVIEW

Project Title: Book Management System **Framework:** Laravel 11 + PHP 8.2 — **Database:** SQLite
Test 1 developed a basic CRUD application for book management with no authentication. Test 2 enhances this with authentication, search functionality, and role-based access control.

2. IMPROVEMENTS MADE SINCE TEST 1

New Features Added

- **Authentication System:** User registration and secure login with email and password. Session-based sessions with role assignment (Admin/User).
- **Search Functionality:** Real-time search by book title and author with clear functionality.
- **Role-Based Access:** Admin users view all books; regular users see only their own books with authorization checks.

Code & Database Improvements

- Separated authentication into dedicated AuthController
- Added `role` column to users table and `user_id` foreign key to books table
- Implemented One-to-Many model relationships (User ↔ Books)
- Added `authorizeBook()` method with ownership verification
- Enhanced validation with email uniqueness and year range checks (1000-2100)

UI/UX Improvements

- Updated header with user welcome message and admin role badge
- Improved navigation with auth state awareness
- Enhanced forms with better validation feedback and color-coded alerts

3. SYSTEM ENHANCEMENTS

Search Implementation: Query builder filters books by title or author using LIKE operator with partial text matching:

Authentication & Authorization: Email-based login with password hashing. Ownership verification prevents users from accessing other users' books:

Validation Enhancements: Email uniqueness, password confirmation, year range (1000-2100), and comprehensive error messages.

4. OBJECT-ORIENTED CONCEPTS USED

Encapsulation

Protected `$fillable` arrays control mass assignment in User and Book models. Private `authorizeBook()` method hides authorization logic from external access. Hidden fields (password, remember_token) protect sensitive data.

Inheritance

User extends Authenticatable (inherits auth methods), Book extends Model (inherits ORM functionality), Controllers extend Controller (inherit middleware). All leverage parent class features without code duplication.

Relationships

One-to-Many: User has many Books; Many-to-One: Book belongs to User. Eloquent relationships simplify data access:

```
// User Model
public function books(): HasMany {
    return $this->hasMany(Book::class);
}
```

5. CHALLENGES & SOLUTIONS

- **SQLite Foreign Keys:** SQLite rejects NOT NULL columns with NULL defaults. Solution: Made `user_id` nullable and enforced authorization in controller.
- **Authorization:** Preventing unauthorized book access. Solution: Implemented `authorizeBook()` method checked on every operation.
- **Session Security:** Session regeneration after login/logout. Solution: Used `$request->session()->regenerate()`.
- **Role Management:** Showing different access levels. Solution: `isAdmin()` helper method with consistent role checks.

6. CURRENT SYSTEM STATUS

What Works Fully: User registration/login with role assignment, book CRUD with ownership tracking, search by title/author, admin access to all books with authorization checks, responsive UI with error handling.

Future Improvements: Email verification, password reset, book categories/genres, ratings/reviews, performance optimization, two-factor authentication, PDF export.

7. REFERENCES

Laravel Official Documentation: <https://laravel.com/docs/11.x> Authentication: <https://laravel.com/docs/11.x/authentication> Eloquent ORM: <https://laravel.com/docs/11.x/eloquent> Database Migrations: <https://laravel.com/docs/11.x/migrations> Validation: <https://laravel.com/docs/11.x/validation>